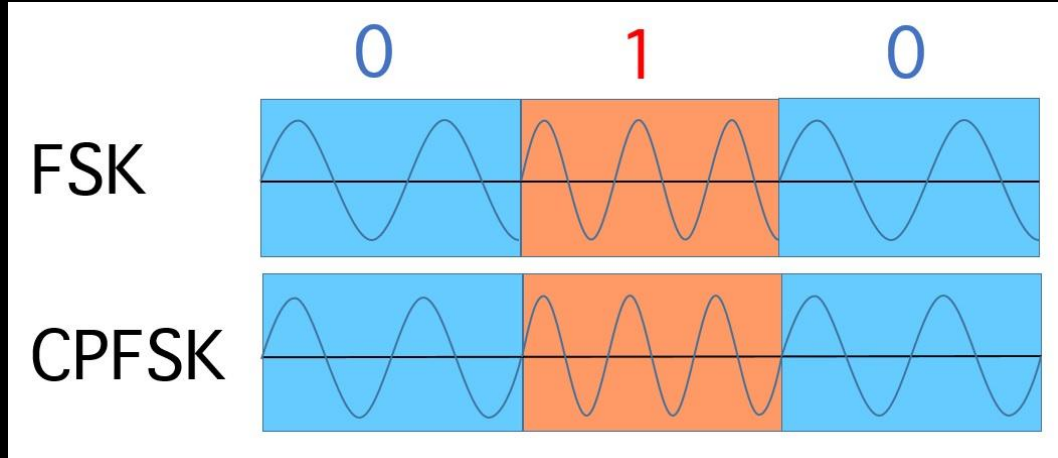


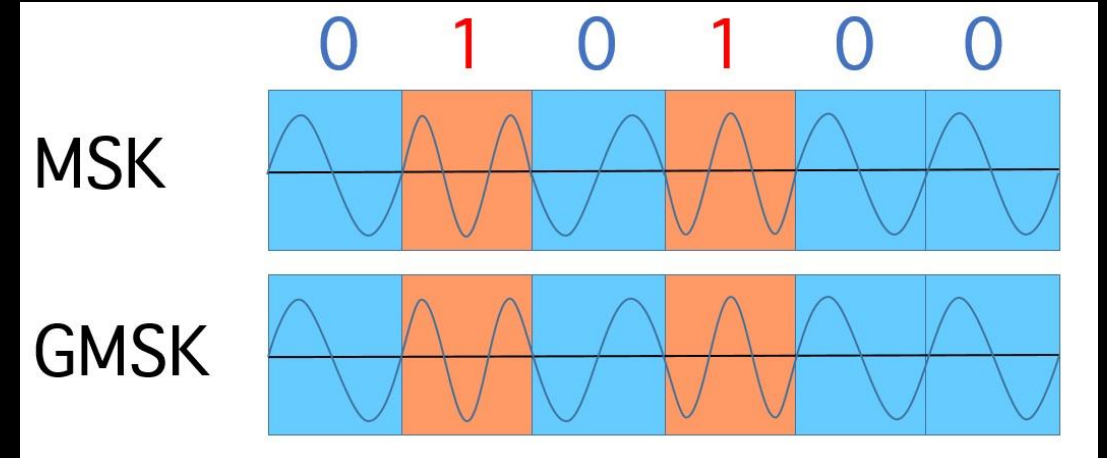
ECE 298A: MSK Modem

Modulation: Varying of one or more properties (amplitude, frequency, phase) of a periodic waveform to transmit information

Minimum-shift keying (MSK): A special case of frequency-shift keying where the phase is continuous (CPFSK) and changes by at most 90° at once



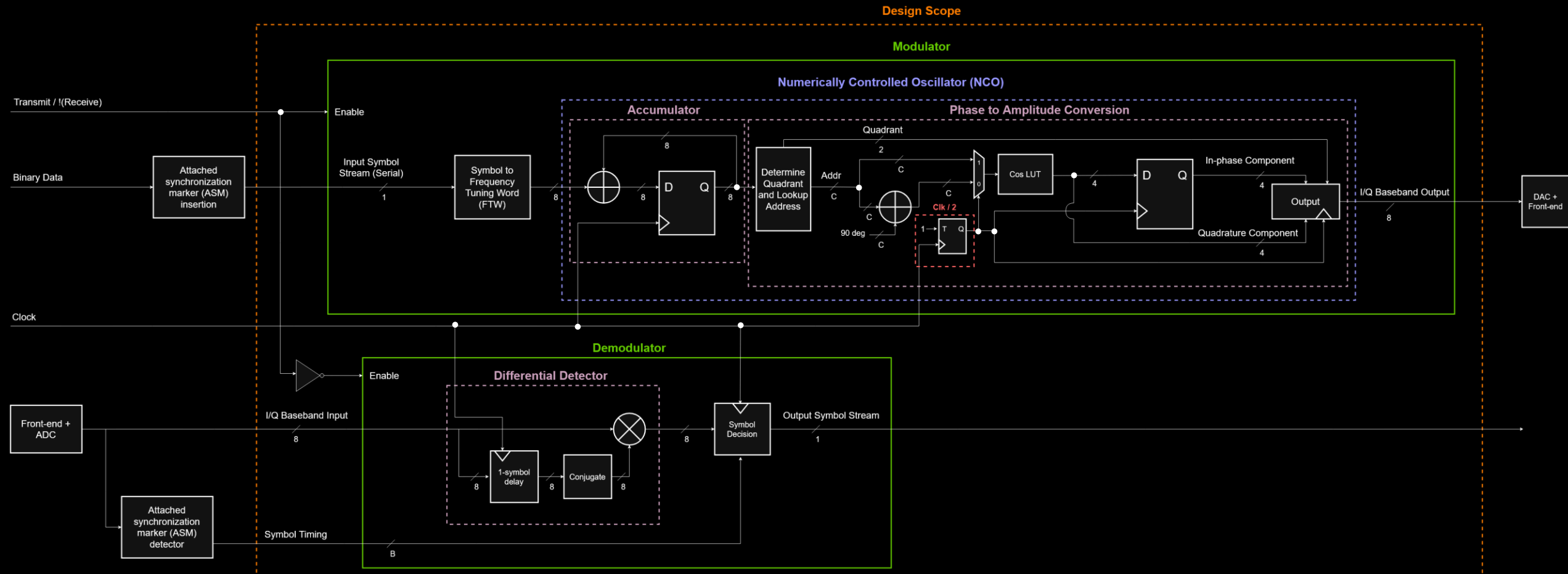
Images by Von Alderamin from scienceblogs.de



Advantages: spectrally efficient compared to regular FSK, resilient to non-linear distortion

Applications: mobile communications (e.g., GSM cell standard), satellite communication

Preliminary Architecture



Specifications + Timeline

Proposed Specifications

Spec	
Symbol rate	200 kbps
Clock	12 MHz
Bit error rate (Receive)	< 0.1% at an Eb/N0 of 10dB
Modulator Input	Serial 1 bit binary
Modulator Output	Parallel 4 bit I/Q (total 8 bits)
Demodulator Input	Parallel 4 bit I/Q (total 8 bits)
Demodulator Output	Serial 1 bit binary
Sample rate	8 samples per symbol
Modulator Latency	6 clock cycles
Demodulator Latency	10 clock cycles

Task	Start Date	End Date	Responsible to
Modulator (transmit) Verilog	Oct 5	Oct 11	Gracia
Demodulator (receive) Verilog	Oct 5	Oct 11	Judy
Modulator Python golden model	Oct 12	Oct 16	Gracia
Demodulator Python golden model	Oct 12	Oct 16	Judy
Cocotb Modulator TB	Oct 17	Oct 23	Gracia
Cocotb Demodulator TB	Oct 17	Oct 23	Judy
Debug + Characterize Modulator RTL with TB results	Oct 24	Nov 22	Judy
Debug + Characterize Demodulator RTL with TB results	Oct 24	Nov 22	Gracia
Parasitic extraction, back annotation, design verification with Open Road	Nov 23	Nov 26	Both
Refine documentation	Nov 27	Dec 3	Both